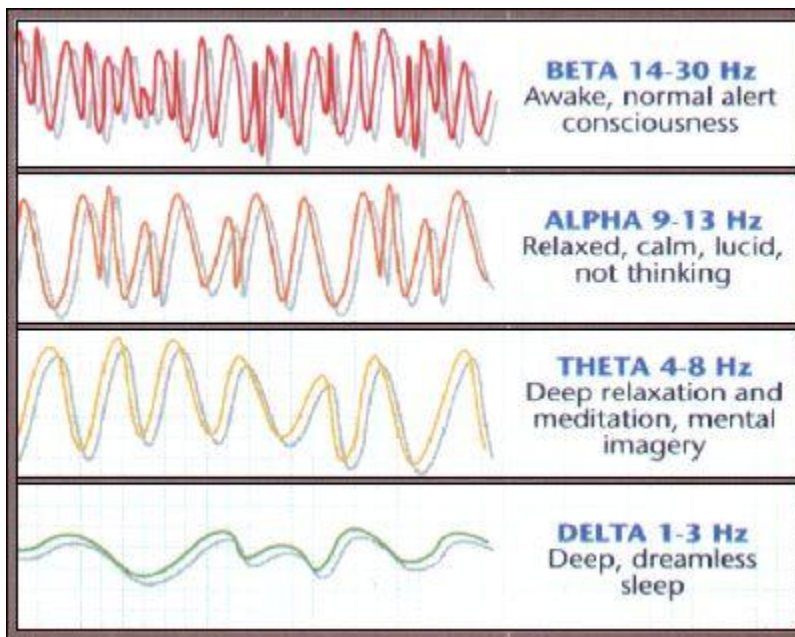


Brain Waves

Notes on Brain Waves, Brain Wave Synchronization and Meditation

The pioneering research on brain waves was conducted in 1908 by Hans Berger, an Austrian doctor, who fell upon alpha waves (8-12 Hz frequency) while studying the phenomenon of telepathy using EEG apparatus. The name alpha is derived from the first letter in the Greek alphabets and those that come after are called beta (12-25Hz). Berger's research showed that brain waves influenced the condition of the brain. As one is alert and vigilant, the brain produces beta waves, but at the same level may exhibit aggressiveness, tension and also stress. However, alpha waves were found to result in general brain relaxation as they calmed down the mind. Later, the Englishmen, Adrian and Matthews discovered delta waves (0.5-4Hz) as W. Grey Walter discovered theta waves (4-12Hz).



Brainwave synchronization

The frequencies of waves created in the human brain came to be known to influence the consciousness of a person.

Brainwave synchronization, also called brainwave entrainment, is concerned with frequency following response, a naturally occurring phenomenon in which similar frequencies tend to synchronize with each other. Brainwave synchronization with binaural beats is said to help synchronize the two hemispheres of the brain, hence the most famous incarnation of this example, Hemi-Sync (by Monroe Institute). Brainwave synchronization has many, early, and often, social manifestations. A drum circle, chanting, as well as staring at a fire, were ways that our ancestors partook in the phenomenon. Crowds at a rock concert, a deep conversation, and being "on the same page" also serve to illustrate how brainwaves can "tune" to certain frequencies. Much is not known about brainwaves and less about their entrainment, and most research is done by private institutions (like the one above) to make and sell these self-help CDs. Brainwave Synchronization is known to exist; but the effectiveness and amount of control is not yet "down to a science".

It is also a private case of functional brain connectivity concept, whereas functional connectivity is defined as the temporal correlation between spatially-remote neurophysiological events, expressed as deviation from statistical independence (temporal correlation) across these events in distributed neuronal groups and areas, which produce the brainwaves.

Brain wave synchronization may be achieved when audio signals are introduced to the brain causing a response directly related to the frequency of the signal introduced, called binaural beats. Two tones close in frequency generate a beat frequency at the difference of the frequencies, which is generally subsonic. For example, a 500 Hz tone and 510 Hz tone will produce a subsonic 10 Hz tone, roughly in the middle of the alpha range. The resulting subsonic tone may affect the state of mind of the subject as with the program I-Doser. The "carrier frequency" (i.e. the 500 Hz in the example above), is also said by some to affect the quality of the transformative experience.

Although some studies have reported that these frequencies do provide help in treating certain medical conditions, there is not a wide acceptance by the medical community to adopt the practice of brainwave synchronization for emotional/mental disorders. The fixed, constant frequency of the synchronization is less helpful than techniques such as classical neurofeedback or learning meditation, which naturally generate brain wave frequencies that differ from person to person and may vary from minute to minute.

Brain wave synchronization is hardly a new technology. Mind machines, audio tapes, and other devices have been floating around for years. Each technique has its strengths and weaknesses. The benefits of these devices are wide in range. Some people use them to relax or fall asleep, while others use the practice of controlling their brain waves to aid in meditation or lucid dreaming.

Our Realities Originate in the Brain

It's necessary to take a step back and look at some basic biological, if not spiritual truths. We create our own realities on every level-- emotional, physical, financial, social, or otherwise.

The Brain is often thought of as a complex, electrically driven organ that works like an incredibly powerful computer. While this is a good analogy, the brain is much

more than that. It actually gives us access to the entire Universe. This vast collection of energy, vibrating and pulsating at various frequencies, is "decoded" for us through the brain. Every aspect of our reality is simulated within the brain to represent what our senses (plus our imagination) know of the Universe around us. The experiences of our senses (color, sense of touch, smell, sound...) are actually an elaborate illusion-- a method by which we have agreed to interpret our surroundings.

The entire Universe is made up of energies that, in and of themselves, are without form or substance. Solid material is, on the sub-atomic level, mostly empty space with particles of energy and electromagnetic fields. It is only the software (i.e. our "belief systems") running in our own brain that gives meaning to this Universe. Our belief system determines how we interpret what we see, hear, taste, smell, and feel. Our emotional reactions and attitudes towards such things as ESP, and other unusual phenomenon are also subject to these learned belief systems.

An important concept to understand is that we are not at the mercy of our current belief system. If we are to live lives of pure satisfaction, if we are to have the experience of manifesting our every desire and to live emotionally satisfying lives, we must understand that there is a higher source of energy over which we have direct control and that we can use this energy to influence how the brain works. Access to this higher energy is made available through specific brain wave patterns. With practice, attaining these brain wave states, while seeking to broaden and change our belief systems, can yield apparent "magical" results. These changes can best be seen in our emotions and our feelings.

What IS Emotion?

Emotion is just one of millions of interpretations and reactions to the universe that we experience every day of our lives. It is no more complicated than that. We decode grass as "green." We might interpret death as "sad" or certain situations as "hopeless." We have learned to interpret certain events and to respond to these events with certain emotions. Many of these emotions, such as fear, are beneficial in events perceived as life threatening, but can be an obstacle to life if they arise without due cause. This sort of "glitch" that impedes our happiness is rooted in our belief system.

Grass is not inherently green by some Universal Law. We, as a species, have simply agreed that the color of grass is a color that we will all call green. As part of a shared "belief system," our brain automatically decodes the color of grass as green. It is an effortless event and one we take for granted as reality.

Imagine that you have just put on a pair of colored sunglasses. At first, everything appears tinted and the normal hues of all colors are seen as different. Eventually, however, we become adjusted to these tints and readily call the grass green, the sky blue and a flower red-- even though these are not the colors our eyes are viewing. This is an example of how strong our belief system can persist as our "reality."

If a number of people, each wearing different colored glasses, can eventually adjust their perception and agree to see the color of grass as "green," despite the fact that each is viewing the grass through a tinted filter, then what is the "truth" about the properties of this color? This leads to a glaring observation about truth: There is no TRUTH about any given situation. We as individuals interpret-- or to put it a better

way-- chose the truth about every situation we encounter. Whether we do it consciously or not is unimportant. It is simply enough to realize-- to FULLY understand-- that everything we consider as truth or fact is no more than our interpretation of an event based largely upon our learned belief system.

To be able to re-learn these emotional responses and change our realities, we must gain direct access to our belief system and make some changes. You can think of it as "upgrading" your belief system to a version of beliefs that will make you a happier person. One of the easiest and most effective ways to do this is in a particular level of consciousness sometimes called the Creative Silence.

Creative Silence refers to a stage of brain activity during which we can most effectively confront and improve our belief system and the visualization of the Universe. It is a stage of brain activity that can be induced and, with instruction, can be used to overcome and undo harmful, deep-rooted beliefs about our own character, abilities and potentials. In technical terms, this is associated with the Theta Waves. Accessing these states can be learned through meditation techniques and biofeedback-- or they can be induced by external stimuli through either audio or visual input. The latter is what we do in brainwave synchronization. The main advantage of using brainwave synchronization over other techniques to achieve this Creative Silence, is that it's effortless. In actuality, brainwave synchronization, as is discussed here, acts as a sort of short-cut to a meditative state.

The practice of meditation is a very powerful tool in the process of any type of reality creation or change.